

QUALITY360 NEXT

STEP 2 PRD

PROCESS MATERIAL QUALIFICATION

Version: 1.0

Target Customer: Vedanta Aluminium – Jharsuguda

Platform: Quality360

Depends On: Step 1 – Incoming Material Inspection

1. BUSINESS OBJECTIVE

Ensure process-critical materials are qualified and released for production consumption before they are used in Carbon Plant, Potline, or Casthouse operations.

Step 1 determines:

"Can we accept this material from the supplier?"

Step 2 determines:

"Can production consume this material safely and efficiently?"

2. BUSINESS CONTEXT

After Incoming Material Inspection approval, selected materials move into production qualification.

Examples:

- Calcined Coke
- Coal Tar Pitch
- Cryolite
- Aluminum Fluoride
- Bath Materials
- Carbon Additives

These materials directly influence:

- Anode Quality
- Potline Efficiency
- Energy Consumption
- Metal Purity
- Process Stability

Incorrect qualification may lead to:

- Process instability
- Excessive energy consumption
- Potline disturbances
- Production losses
- Quality deviations

3. MODULE NAME

Process Material Qualification

Menu Location:

Quality Operations → Process Material Qualification

4. PRIMARY USERS

Process Engineer

Responsibilities:

- Initiate qualification request
- Review process suitability

Lab Analyst

Responsibilities:

- Perform tests
 - Import instrument results
 - Validate test completion
-

QA Engineer

Responsibilities:

- Review results
 - Evaluate readiness
-

QA Manager

Responsibilities:

- Release
 - Hold
 - Reject
-

Viewer

Responsibilities:

- Read-only access
-

5. BUSINESS FLOW

Material Approved in Step 1 ↓ Qualification Request Created ↓ Sample Generated ↓ Tests Assigned ↓ Results Captured ↓ Process Readiness Evaluation ↓ QA Review ↓ Release / Hold / Reject ↓ Available For Production Consumption

6. PAGE STRUCTURE

Page 1: Qualification Queue

Page 2: Process Material Qualification Workbench

Reuse all framework components from Step 1.

Do not redesign platform architecture.

7. PAGE 1

QUALIFICATION QUEUE

Purpose:

Display all qualification requests.

Columns:

Qualification No Material Batch No Consumption Area Status Risk Assigned User

Actions:

Create Qualification Open Workbench Clone Cancel View History

Filters:

Material Area Status Date Range

8. PAGE 2

PROCESS MATERIAL QUALIFICATION WORKBENCH

Main execution screen.

Reuse Step 1 layout.

Sections:

Header Workflow Timeline Material Overview Sample Management Test Workspace Activity Feed Quality
Insights Approval Center Audit Trail

9. HEADER

Display:

Qualification Number Material Consumption Area Batch Number Status

Example:

PMQ-2026-001245

Calcined Coke

Carbon Plant

CC-2026-015

Under Review

10. WORKFLOW TIMELINE

Stages:

Request ↓ Sample ↓ Testing ↓ Validation ↓ Review ↓ Release

Visual progress indicator mandatory.

11. MATERIAL OVERVIEW

Fields:

Material Batch Number Supplier Consumption Area Quantity Request Date Requested By

Actions:

Edit View History

12. MATERIAL MASTER DATA

Supported Demo Materials:

Calcined Coke

Coal Tar Pitch

Cryolite

Aluminum Fluoride

Bath Material

Carbon Additive

Pet Coke

13. CONSUMPTION AREAS

Carbon Plant

Potline

Casthouse

R&D

14. SAMPLE MANAGEMENT

Functions:

Create Sample

Edit Sample

Recollect Sample

View Sample History

Example Sample ID:

PMQS-001245

Processing Animation:

Creating Sample... Assigning Workflow... Generating Qualification Record... Completed Successfully

15. TEST WORKSPACE

Purpose:

Capture qualification results.

Supported Sources:

Instrument Import

Manual Entry

File Upload

16. TEST MATRIX

Calcined Coke

Tests:

Sulphur

Moisture

Density

Air Permeability

Electrical Resistance

Coal Tar Pitch

Tests:

Softening Point

Density

Viscosity

Cryolite

Tests:

Fluoride Content

Composition

Moisture

Bath Material

Tests:

Chemical Composition

Moisture

Phase Analysis

17. INSTRUMENT INTEGRATION

Reuse Step 1 Integration Framework.

Supported Instruments:

XRF

XRD

Carbon Sulphur Analyzer

Moisture Analyzer

Pycnometer

Air Permeability Analyzer

Electrical Resistance Analyzer

Softening Point Tester

18. INSTRUMENT IMPORT EXPERIENCE

When Import Result is clicked:

Step 1

Connecting to Instrument...

Progress 10%

Step 2

Verifying Communication...

Progress 25%

Step 3

Reading Batch Data...

Progress 45%

Step 4

Parsing Results...

Progress 65%

Step 5

Validating Structure...

Progress 85%

Step 6

Import Successful

Progress 100%

Total Duration: 5 seconds

19. MOCK RESULTS

Calcined Coke

Sulphur = 1.25 %

Moisture = 0.35 %

Density = 2.08

Air Permeability = 14.5

Electrical Resistance = 52

20. QUALITY INSIGHTS

Replace Step 1 Supplier Health.

New Hero KPI:

Process Readiness

Display:

Process Readiness Score

Risk Level

Recommended Action

Observations

Historical Comparison

Tests Complete

21. PROCESS READINESS SCORE

Calculated Using:

Test Completion

Specification Compliance

Historical Stability

Deviation Count

Risk Indicators

Display:

0 - 100

Example:

91 / 100

22. RECOMMENDATION ENGINE

Possible Outputs:

Release

Review

Hold

Reject

Example:

Recommended Action

RELEASE TO CARBON PLANT

Reason:

All critical parameters compliant.

Historical variation low.

No process risk identified.

23. OBSERVATIONS ENGINE

Examples:

Sulphur within acceptable range.

Moisture stable compared to previous batches.

Density compliant with Carbon Plant requirement.

No abnormal variation detected.

24. HISTORICAL COMPARISON

Example:

Current Sulphur = 1.25 %

Previous Average = 1.18 %

Difference = +0.07 %

Display trend indicator.

25. ACTIVITY FEED

Reuse Step 1 framework.

Events:

Qualification Created

Sample Created

Result Imported

Validation Completed

Readiness Updated

Release Approved

Newest First

Real-Time Updates

26. AUDIT TRAIL

Reuse Step 1 framework.

Track:

User

Timestamp

Action

Previous Value

New Value

Source

Examples:

Result Modified

Approval Completed

Release Override

Comment Added

27. APPROVAL CENTER

Actions:

Release

Hold

Reject

Reason mandatory for Hold and Reject.

Success Messages:

Material released successfully.

Material placed on hold.

Material rejected for process use.

28. NOTIFICATIONS

Every action generates feedback.

Examples:

Qualification created successfully.

Sample generated successfully.

Results imported successfully.

Process readiness recalculated.

Material released to Carbon Plant.

29. ROLE PERMISSIONS

Process Engineer

Create Qualification View Results

Lab Analyst

Import Results Manual Entry Upload Results

QA Engineer

Review Recommend

QA Manager

Release Hold Reject Override

30. DEMO SCENARIO

Qualification Number:

PMQ-2026-001245

Material:

Calcined Coke

Consumption Area:

Carbon Plant

Batch:

CC-2026-015

Results:

Sulphur 1.25 %

Moisture 0.35 %

Density 2.08

Air Permeability 14.5

Electrical Resistance 52

Process Readiness:

91 / 100

Risk:

LOW

Recommendation:

RELEASE TO CARBON PLANT

31. REUSE REQUIREMENT

Mandatory:

Reuse all platform components from Step 1:

Workbench Framework

Instrument Simulation

Activity Feed

Audit Trail

Notifications

Approval Framework

Quality Insights Framework

No duplicate architecture.

Only business logic changes.

SUCCESS CRITERIA

A user should be able to:

Create Qualification

Generate Sample

Import Instrument Results

Observe Process Readiness Updates

Review Insights

Approve Release

See Audit Trail

Receive Notifications

All within a single integrated workbench experience.